IMAGE PRIVACY LEAKAGE IN AI: A STUDY OF GRADIENT RECONSTRUCTION AND FACE PIXELATION

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Abstract

This project investigates privacy leakage in image-based AI systems, with a focus on gradient leakage and image reconstruction. The study began with a literature review of privacy attacks and existing defences, then moved into a small proof-of-concept prototype. Two pipelines were developed: a baseline leakage pipeline and a defended pipeline. In the baseline case, gradients were used to reconstruct an approximate image. In the defended case, face pixelation was applied before training so that the reconstructed output became less clear. The results showed that gradients can leak meaningful visual information and that the proposed defence can reduce how revealing the reconstruction becomes.

Introduction

Image-based AI systems are widely used for editing, generating, and analysing visual content. However, these systems also raise privacy concerns because images can contain sensitive personal information. A key issue is gradient leakage, where gradients shared during training may reveal enough information for an attacker to reconstruct the original image. Related research has shown that privacy risks can also continue after training through model inversion attacks. This project focuses on gradient leakage as the main practical threat and studies whether a simple image-level defence can reduce the risk.

Project Idea Objectives

This project investigates privacy leakage in image-based AI systems, with a particular focus on gradient leakage and image reconstruction. The study combines a literature-based review with a small proof-of-concept prototype to show how leaked gradients can reveal sensitive visual information. Two pipelines were developed: a baseline leakage pipeline and a defended pipeline using face pixelation during training. The aim was to compare both cases and evaluate whether the defence could reduce how clear and revealing the reconstructed output became.

1. Investigate privacy leakage in image-based AI systems.
2. Study gradient leakage and model inversion as privacy threats
3. Review existing defences from the literature.
4. Build a baseline proof-of-concept leakage model..
5. Build a defended version using face pixelation during training.
6. Compare both pipelines visually and numerically.
7. Evaluate the results using MSE, PSNR, and SSIM.

Project Lifecycle

Literature Review
Review of the main research on gradient leakage, model inversion, and privacy defences in image-based AI systems.

Defence Review and Planning
Theoretical review of existing defence strategies and refinement of the practical defence direction for the project.

Prototype Development
Development of a small proof-of-concept model using Python and Google Colab, including a baseline leakage pipeline and a defended pipeline.

Testing and Evaluation
Testing of both pipelines using reconstructed outputs and image quality metrics such as MSE, PSNR, and SSIM.

Final Report and Poster Preparation
Writing up the project findings, discussing limitations and future work, and preparing the final poster and presentation materials.

Methodology

This project combined a literature review with a small proof-of-concept prototype. An input image was passed through a CNN model to produce gradients during training. These gradients were treated as leaked information. A dummy image was then created and updated repeatedly by matching its dummy gradients to the leaked gradients. As the values became closer, the dummy image gradually formed an approximate reconstruction of the original image. The same process was tested in a baseline pipeline and a defended pipeline using face pixelation, and the outputs were compared using MSE, PSNR, and SSIM.

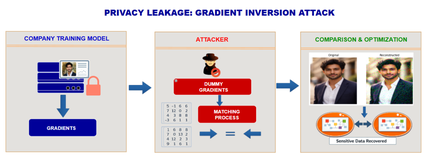


Figure 1: System architecture of the baseline and defended pipelines.

Product

The prototype was built around two pipelines:

* Baseline pipeline

Input image → preprocessing → CNN model → forward pass → loss calculation → backward pass → gradients saved → gradient inversion → reconstructed image

* Defended pipeline

Input image → face pixelation → preprocessing → CNN model → forward pass → loss calculation → backward pass → defended gradients saved → gradient inversion → reconstructed image

The aim was to compare whether the defence reduced the clarity and recognisability of the leaked reconstruction.

Results

The baseline pipeline successfully reconstructed an approximate image from leaked gradients. The output retained visible facial structure and general visual information, showing that the leakage process worked.

METRICS	BASELINE	DEFENDED
MSE	0.042849	0.059932
PSNR	13.68db	12.22db
SSIM	0.4148	0.2569

Table 1:

The defended pipeline also produced a reconstruction, but the output was more blurred and less detailed. This suggested that the proposed defence reduced how much useful information could be recovered.

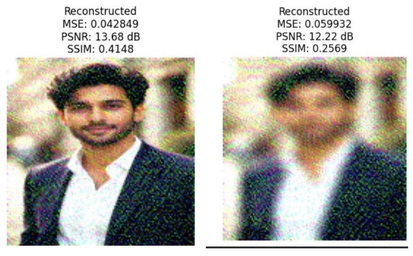


Figure 2: Baseline and defended outputs with metrics.

Results may vary slightly between runs because each reconstruction starts from scratch using the collected gradients, rather than reusing saved reconstruction data.

Evaluation

The project met its main aim by showing that gradients can leak image information and that this can be demonstrated through a small prototype. The baseline pipeline produced a more recognisable reconstruction, while the defended pipeline produced a blurrier and less detailed result. This suggests that the proposed defence reduced how revealing the leaked output became. At the same time, the project had clear limitations, as it used a small CNN-based proof of concept rather than a larger architecture or a full federated learning environment. Even with these limits, the project was valuable because it combined theory with practical testing and showed the privacy risk in a clear and manageable way.

Conclusions

- Gradients can leak meaningful image information in AI systems.
- The baseline pipeline showed that an approximate image could be reconstructed from leaked gradients.
- The defended pipeline showed that face pixelation reduced how clear and revealing the reconstruction became.
- Overall, the project gave a clearer understanding of privacy risks in image-based AI and how their impact may be reduced.

Limitations Further Work

This project was developed as a small proof of concept, so it used a simple CNN-based architecture rather than a larger model such as ResNet. It was also tested in a simplified setting rather than in a full federated learning environment. In addition, only one practical defence was implemented in the prototype, even though the literature discusses several other privacy-preserving methods. Future work could extend this by testing larger architectures, more realistic federated learning settings, and additional defences such as Differential Privacy or Variational Bottleneck methods.

References

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[2] J. Geiping, H. Bauermeister, H. Dröge, and M. Moeller. Inverting Gradients: How Easy Is It to Break Privacy in Federated Learning? In *Advances in Neural Information Processing Systems*, volume 33, 2020.